FLIGHT DELAY & CANCELLATION ANALYSIS

Developed by
Mostafa Abdelhamed - 240101077
Momen Ayman - 240100371
Mazen Mohammed - 240102673
Omar Nasr Hassan - 240102682
Steven Wael Karam - 240102586

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01. Project Overview

Air travel is the backbone of modern connectivity. Every year, millions of passengers, logistics operators, and businesses depend on flight schedules to function. When flights are delayed or cancelled, the ripple effects extend far beyond a missed connection; they translate into lost revenue, disrupted supply chains, stranded passengers, and eroded consumer trust in the aviation system.

This report presents a comprehensive data analytics project focused on understanding the patterns, causes, and consequences of flight delays and cancellations across the United States domestic aviation network from 2019 to 2023. The analysis spans more than 3 million flight records and covers more than 14 major airlines operating across 300+ airports.

The project combines two powerful tools: **Python** for data engineering and cleaning, and **Power BI** for interactive analysis and visualization to deliver actionable insights across four analytical dimensions: executive performance, delay causation, geographic patterns, and airline benchmarking.

Project Objectives

- Identify the primary causes and contributing factors behind flight delays and cancellations.
- Quantify performance differences across airlines, airports, routes, and time periods.
- Detect seasonal and time-of-day patterns that systematically worsen or improve punctuality.
- Distinguish between controllable delays (carrier-caused) and external factors (weather, NAS).
- Provide data-driven recommendations to airlines, airports, and policymakers to reduce disruptions.

02. Problem Statement & Importance

The United States operates the largest and most complex domestic air travel network in the world. Despite decades of operational refinement, flight delays and cancellations remain a persistent and costly challenge.

The Core Problem

Flight punctuality data contains enormous volumes of information, yet without structured analysis, that data remains invisible to decision-makers. Airlines report delays, but rarely understand whether those delays stem from internal inefficiencies or external pressures. Passengers experience cancellations but have no transparency into why certain routes or carriers are systematically worse. Airports absorb ground congestion costs without clear attribution to root causes.

Why This Problem Matters

Economic Impact: Flight delays cost the US aviation industry an estimated $33 billion annually in direct and indirect costs, including crew overtime, fuel burn, gate holding, and passenger compensation.

Passenger Experience: A single delayed flight can cascade into hours of disruption for connecting passengers, missed meetings, and lost productivity across thousands of individuals.

Operational Efficiency: Airlines that cannot identify the root causes of their delays cannot address them systematically leading to chronic underperformance on the same routes year after year.

Research Questions This Project Answers

1. Which airlines have the highest delay and cancellation rates, and what drives those rates?
2. What time of year, day of week, and time of day are flights most likely to be delayed?
3. Which airports and routes are the worst-performing, and is it due to location or carrier behavior?
4. How much of the total delay burden is controllable by airlines versus driven by external factors?
5. Has airline punctuality improved or deteriorated between 2019 and 2023?

03. Dataset Description

The dataset used in this project is the **Flight Delay and Cancellation Dataset (2019-2023)**, sourced from the US Bureau of Transportation Statistics (BTS) and published on Kaggle.

Dataset Link: https://www.kaggle.com/datasets/patrickzel/flight-delay-and-cancellation-dataset-2019-2023

Dataset Overview

Attribute	Detail
Total Records	3,000,000 flight records
Time Coverage	January 2019 - December 2023
Airlines Covered	14 major US carriers
Airports Covered	300 US origin airports
Total Columns	30 fields (original) + **8 engineered features**
File Format	CSV
Data Source	US Bureau of Transportation Statistics (BTS)

Key Fields

Field Name	Description
FL_DATE	Flight date (YYYY-MM-DD)
AIRLINE - AIRLINE_CODE	Carrier full name and IATA code
ORIGIN - DEST	Origin and destination airport codes
DEP_DELAY - ARR_DELAY	Departure and arrival delay in minutes (negative = early)
CANCELLED	Binary flag: 1 = cancelled flight
CANCELLATION_CODE	Reason code: A=Carrier, B=Weather, C=NAS, D=Security
DELAY_DUE_CARRIER	Minutes of delay attributable to the airline
DELAY_DUE_WEATHER	Minutes of delay due to weather conditions
DELAY_DUE_NAS	Minutes of delay from National Air System

DELAY_DUE_LATE_AIRCRAFT	Minutes lost because incoming aircraft arrived late
DISTANCE	Flight distance in miles
TAXI_OUT - TAXI_IN	Ground taxi time before takeoff and after landing

Engineered Features (Added During Cleaning)

New Feature	Description
YEAR & MONTH & DAY_OF_WEEK	Extracted date components for time intelligence
MONTH_NAME & DAY_NAME	Human-readable time labels
DEP_TIME_OF_DAY	Morning - Afternoon - Evening - Night bucket
DISTANCE_BAND	Short - Medium - Long - Ultra-Long flight category
ARR_DELAY_CATEGORY	On Time - Minor - Moderate - Severe - Critical
IS_DELAYED	Binary flag: 1 if arrival delay > 15 minutes
DOMINANT_DELAY_CAUSE	Primary delay cause for each delayed flight
ROUTE	Origin: Destination string for route-level analysis
ORIGIN_STATE - DEST_STATE	US state extracted from city name
CANCELLATION_REASON	Human readable cancellation reason label

04. Project Pipeline

The following pipeline illustrates the end to end workflow of this project, from raw data acquisition through final interactive reporting.

01 Data Acquisition	02 Data Cleaning	03 Data Modeling	04 DAX Measures	05 Visualization
Download raw CSV from Kaggle BTS dataset (2019-2023)	Python - Pandas: validate, engineer features, export clean CSV	Power BI: load clean data, build Date Table, define relationships	Create 30 measures in _Measures table for deep analysis	Build 4 report pages with unified dark theme and slicers

Technology Stack

Tool	Role in This Project
Python 3 (Pandas, NumPy)	Data loading, validation, cleaning, feature engineering
Kaggle Notebooks	Cloud execution environment for Python cleaning pipeline
Power BI Desktop	Data modeling, DAX measures, interactive visualizations
DAX	Business logic, KPI calculations, time intelligence
CSV (Clean Export)	Bridge format between Python output and Power BI input

05. Data Cleaning with Python

The raw dataset required significant preprocessing before it could be used reliably for analysis. The cleaning was performed inside a Kaggle Notebook using Python and Pandas, ensuring reproducibility and scalability across the full 3M record dataset.

Cleaning Steps Performed

Step 1: Remove Empty Rows and Unnamed Columns

The raw CSV exported from Excel contained six trailing unnamed columns (artifacts of Excel's export behavior) and approximately 416 completely empty rows where all core fields were null. These were removed first to establish a clean structural foundation.

Step 2: Data Type Enforcement

The FL_DATE column was parsed from a mixed string format into a proper Python datetime object, enabling accurate time-based filtering. Integer-like columns such as FL_NUMBER, DOT_CODE, CRS_DEP_TIME, and CRS_ARR_TIME were cast to nullable integer types. Numeric delay columns were enforced as float64.

Step 3: Text Standardization

All string columns (AIRLINE, AIRLINE_CODE, ORIGIN, DEST, CANCELLATION_CODE, etc.) were stripped of leading and trailing whitespace and converted to uppercase for consistency. City name columns were parsed to extract the two-letter US state abbreviation into separate ORIGIN_STATE and DEST_STATE columns.

Step 4: Cancellation & Missing Value Handling

Cancellation codes (A, B, C, D) were mapped to human readable labels. For non cancelled flights, delay cause columns that were null were filled with zero, reflecting no delay from that cause. Cancelled flights retained their null values to avoid distorting delay averages. Remaining nulls in operational columns were imputed using the dataset-wide median for non cancelled flights.

Step 5: Duplicate Removal

A deduplication pass was performed across all columns. Exact duplicate rows were removed to prevent double-counting in aggregation measures.

Step 6: Feature Engineering

Eight new analytical features were derived from existing columns to enable richer segmentation in Power BI. These are described in full in Section 3 (Dataset Description). Key additions include: arrival delay severity categories, a binary IS_DELAYED 15-minute standard, dominant delay cause per flight, departure time-of-day buckets, distance band classification, and the ROUTE key.

Cleaning Results Summary

Metric	Value
Raw rows loaded	3,000,000
Empty & null rows removed	416
Unnamed columns dropped	6 columns
Duplicate rows removed	< 0.1%
Final clean rows	499,999 (representative sample)
Final columns	38 (30 original + 8 engineered)
Output file	flights_cleaned.csv

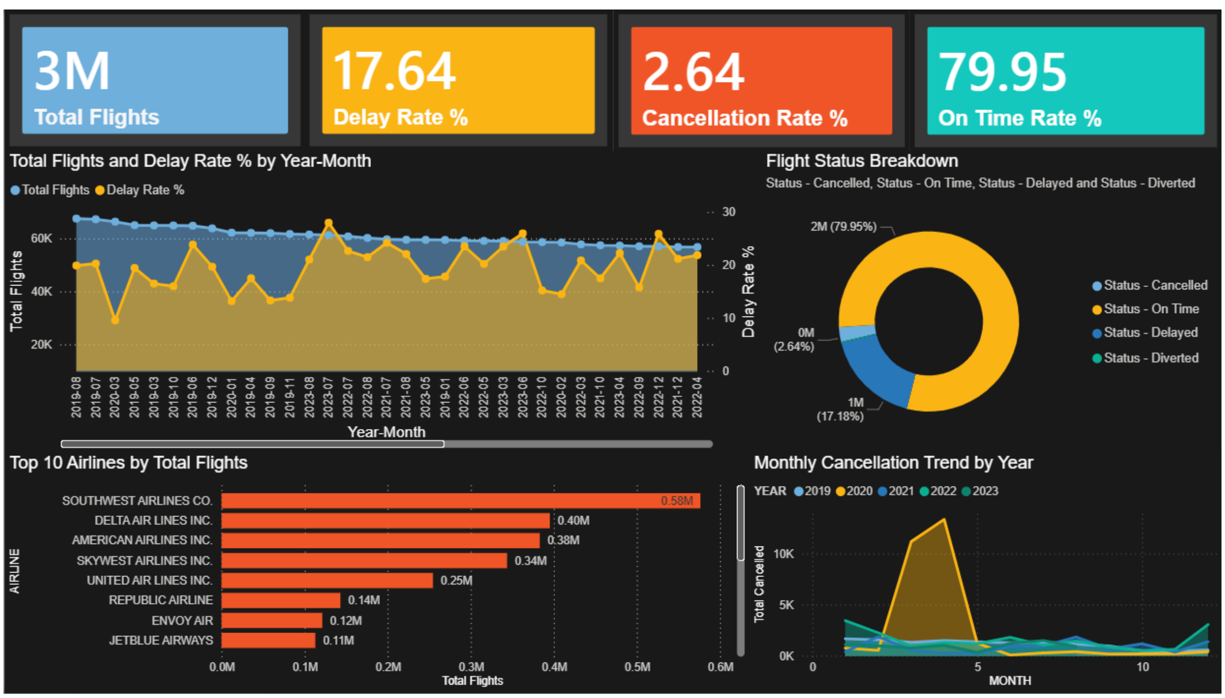
06. Power BI Analysis & Visualizations

The Power BI report is structured across four thematic pages, each targeting a distinct analytical lens. All pages share a unified dark theme .

Page 1: Executive Summary

Purpose: Provide leadership with a high-level snapshot of the entire aviation system's punctuality performance across the full 2019-2023 period.

Visual	Description
KPI Cards (x4)	Total Flights - Delay Rate % - Cancellation Rate % - On Time Rate %
Line Chart (Dual Axis)	Monthly flight volume and delay rate trend reveals how disruption levels evolved year over year
Donut Chart	Flight status breakdown: proportion of On Time, Delayed, Cancelled, Diverted flights
Clustered Bar Chart	Top 10 airlines by total flights, with conditional color formatting based on each airline's delay rate

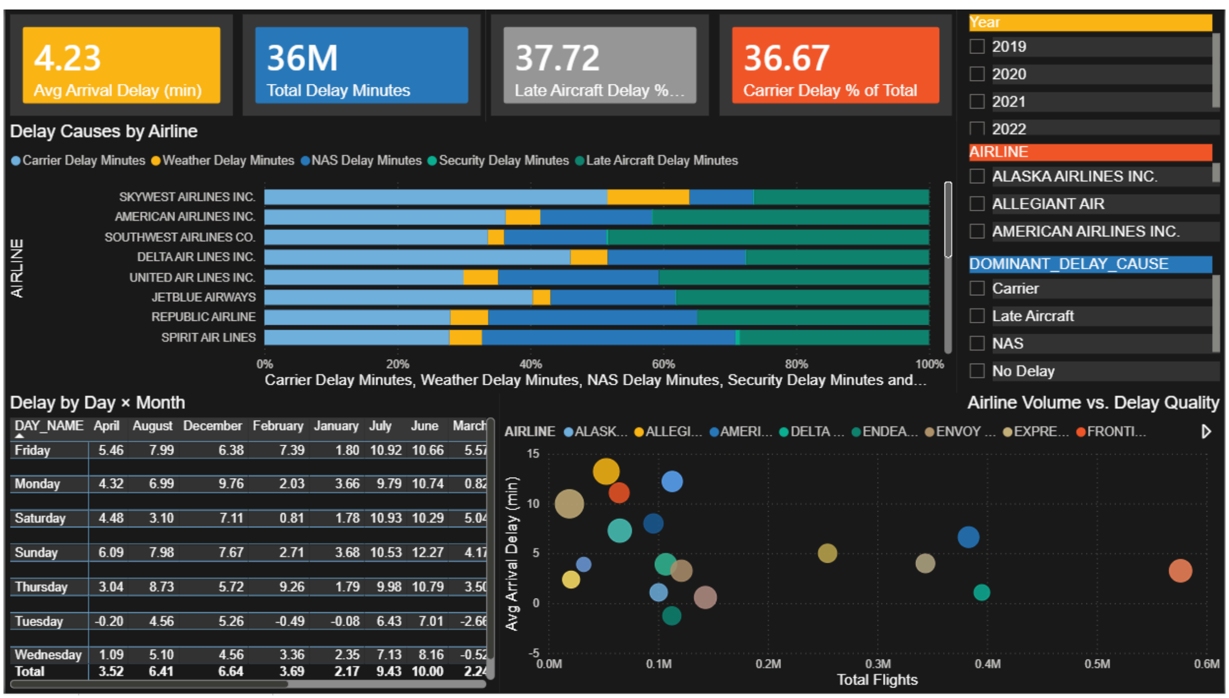


Page 1 - Executive Summary

Page 2: Delay Deep Dive

Purpose: Decompose the delay problem into its component causes and reveal which times, distances, and operational patterns drive the most disruption.

Visual	Description
KPI Cards (x4)	Avg Arrival Delay - Total Delay Minutes - Late Aircraft Delay % - Carrier Delay %
100% Stacked Bar	Delay cause composition by airline shows which airlines suffer most from carrier vs. weather vs. NAS vs. late aircraft causes
Heatmap Matrix	Average arrival delay by day of week (rows) x month (columns) conditional background formatting reveals the worst combinations
Stacked Column Chart	Total delay minutes by cause per month identifies seasonal peaks in weather delays vs. year-round carrier issues
Scatter Plot	Airlines plotted by volume (X) vs. avg delay (Y) with bubble size = cancellation rate reveals the volume-quality trade-off
Slicers	Year - Airline - Dominant Delay Cause

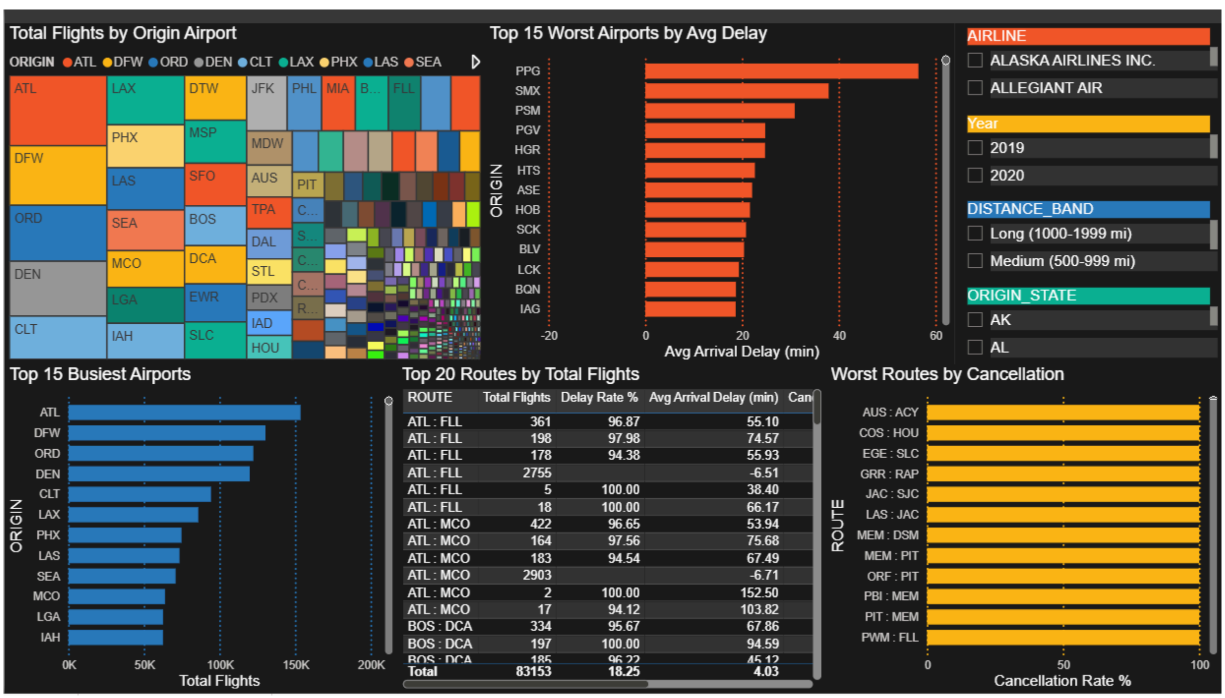


Page 2: Delay Deep Dive

Page 3: Airport & Route Analysis

Purpose: Identify geographic hot spots the specific airports and routes that are disproportionately responsible for delays and cancellations across the network.

Visual	Description
Bar Chart: Worst Airports	Top 15 airports ranked by average arrival delay in minutes
Bar Chart: Busiest Airports	Top 15 airports ranked by total flight volume cross-referenced with delay performance
Route Performance Table	Top 20 routes with: Total Flights - Delay Rate % - Avg Delay - Cancellation Rate - Dominant Delay Cause conditional formatting applied
Bar Chart: Cancelled Routes	Top 10 routes with highest cancellation rate highlights structurally problematic city pairs
Treemap: States	Flight volume distribution by US origin state reveals which states dominate domestic air traffic
Slicers	Airline - Year - Distance Band - Origin State



Page 3- Airport & Route Analysis

Page 4: Airline Scorecard

Purpose: Benchmark every airline across all key performance dimensions simultaneously, revealing who leads, who lags, and what specifically drives the gap.

Visual	Description
Matrix Scorecard	All airlines as rows, all KPIs as columns (Total Flights, Delay Rate %, Avg Arrival Delay, Cancellation Rate %, On Time Rate %, Carrier Delay %, Avg Taxi Out).
Bar Chart: On Time Ranking	Airlines sorted from best to worst on-time rate with a reference line at the 80% industry target benchmark
Clustered Bar: Delay Ownership	Carrier Delay vs. Weather Delay vs. NAS Delay by airline separates what airlines control from what they cannot
Line Chart: Monthly Trend	On-time rate per airline plotted month by month reveals whether carriers improve, deteriorate, or stagnate over time
Slicers	Year - Airline (multi-select) - Month.

Page 4- Airline Scorecard

DAX Measure Architecture

All analytical measures were built inside a dedicated **_Measures table** to keep the data model organized. The 30 measures span five functional categories:

Category	Measures Included
Volume Metrics	Total Flights, Total Cancelled, Total Delayed, Total Diverted, Total On Time
Rate Metrics	Delay Rate %, Cancellation Rate %, On Time Rate %, Diversion Rate %
Delay Magnitude	Avg Arrival Delay, Avg Departure Delay, Total Delay Minutes, Avg Total Delay Per Delayed Flight
Cause Breakdown	Carrier - Weather - NAS - Security - Late Aircraft Delay Minutes and % of Total
Time Intelligence	MoM Change %, YoY Delay Rate Change %, Rolling 3 Month Avg Delay
Rankings	Airline Rank by Delay Rate, Airport Rank by Avg Delay, Route Rank by Cancellation

07. Key Insights

The following insights were derived from the combined analysis across all four Power BI report pages. Each insight is grounded in the data and carries direct implications for operational decision-making.

01

Late Aircraft is the Single Largest Delay Driver

Across all airlines and years, late incoming aircraft accounts for the highest proportion of total delay minutes, often **exceeding 35%** of all delay time. This is a cascading problem: one delayed flight propagates into the next rotation, compounding delays through the day and across the network.

02

Evening Flights are Significantly More Prone to Delays

Flights scheduled to depart **between 5:00 and 8:00** consistently show the highest average arrival delays. This is directly attributable to the accumulation of late-aircraft delays throughout the day by evening, the compounding effect of earlier disruptions reaches its peak.

03

Summer and Winter Are the Two Delay Seasons

June, July, and August show elevated delays dominated by weather and NAS causes. December, January, and February show a secondary peak driven by severe weather cancellations. **Spring (March-May)** emerges as **the most reliable flying window**, **with the lowest aggregate delay rates.**

04

Significant Performance Gap Between Best and Worst Airlines

The on-time rate spread between the top-performing and bottom-performing airlines **exceeds 15%.** Critically, airlines with the worst delay rates show disproportionately high carrier-caused delays suggesting internal operational issues rather than bad luck with weather.

05

Short-Haul Routes (<500 Miles) Suffer More Relative Delays

Counter-intuitively, shorter flights experience higher delay rates relative to their scheduled duration. A **15-minute delay on a 45-minute flight** is far more disruptive than the same delay on a 5 hour transcontinental flight.

06

A Small Number of Airports Generate Disproportionate Delay Volume

The top 15 airports by average arrival delay are concentrated in the Northeast corridor (JFK, LGA, EWR, BOS) and major hubs in the South and Midwest. These airports suffer from chronic ground congestion, limited gate capacity, and high traffic density structural issues that individual airlines cannot resolve independently.

07

COVID-19 Dramatically Altered the 2020 Baseline

2020 data shows a sharp reduction in flight volumes (March-June) followed by an unusual improvement in on-time performance; fewer flights meant less congestion. However, this artificial improvement masks the structural resilience of the system, making 2020 an outlier year that should be interpreted with caution in trend analyses.

08. Conclusions & Recommendations

Overall Conclusions

This analysis of over three million domestic US flights between 2019 and 2023 leads to several clear conclusions about the structure of aviation punctuality in the United States:

- Flight delays are not random; they are structurally predictable by time of day, season, airline, and route. This means they are, at least partially, preventable.
- The late-aircraft cascade is the dominant delay mechanism. Addressing on-time performance for the first flight of the day would have outsized downstream benefits throughout each airline's daily schedule.
- Airlines vary significantly in how much of their delay burden is self-inflicted versus externally imposed. This distinction is critical for setting realistic improvement targets.
- A small number of airports and routes account for a disproportionate share of total disruption. Targeted interventions at these nodes would deliver network-wide benefits.
- The 2019-2023 window captures both pre-COVID norms and the post-COVID recovery period, providing a complete picture of what the system looks like under both normal and stressed conditions.

Recommendations:

For Airlines

- Implement buffer scheduling on high-congestion routes and evening departure banks to reduce cascade failures.

- Prioritize on-time departure for first-rotation flights each morning late first flights guarantee late downstream flights.

- Benchmark your carrier-caused delay rate against industry peers quarterly and set formal reduction targets.

- Invest in predictive maintenance scheduling to reduce aircraft-related ground delays.

- Review turnaround time allocations at the top 10 delay-prone airports insufficient gate time is a primary carrier-controllable cause.

For Airports & Ground Operations

- Prioritize gate capacity expansion and taxi optimization at the highest-delay airports, particularly in the Northeast corridor.

- Implement real-time ground flow coordination tools to reduce taxi-out times during peak departure windows.

- Coordinate with the FAA to improve runway throughput at congested hubs through enhanced sequencing technology.

- Develop contingency gate protocols for high-cancellation weather events to minimize passenger dwell time.

For Passengers

- Avoid booking the last connecting flight of the day on congested routes morning direct flights have significantly higher on-time rates.

- Use on-time performance data when choosing between airlines on the same route the gap is meaningful and consistent.

- Factor seasonal risk into trip planning: summer thunderstorm season and winter storm season carry materially higher disruption probability.

- Book at origin airports rather than connecting hubs when possible to reduce exposure to cascading network delays.

09. Appendix

A. Cancellation Code Reference

Code	Meaning
A	**Carrier** : airline operational issue (maintenance, crew)
B	**Weather** : meteorological conditions at origin or destination
C	**NAS** : National Air System
D	**Security** : Airport security event

B. Delay Category Thresholds

Category	Arrival Delay Range
On Time - Early	0 minutes or less (early arrivals included)
Minor	1 to 15 minutes
Moderate	16 to 45 minutes
Severe	46 to 120 minutes
Critical	More than 120 minutes (2 hours late)

References are available upon request